RE.Ai Solution

Intelligent Virtual Staging through Dual-Pass Computer Vision and Generative AI Synthesis.

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Project Tier: Tier 2 (Bonus Points Applied)

ITAI 1378: Computer Vision Course

Problem & Motivation

The Vacancy Gap

Vacant properties stay on the market 35% longer and sell for 10% less than staged ones. Buyers struggle to visualize spatial layout without context.

Cost Inefficiency

Physical staging costs between $2,000 to $5,000 per month. Traditional virtual staging requires manual labor and hours of graphic design work.

The Solution

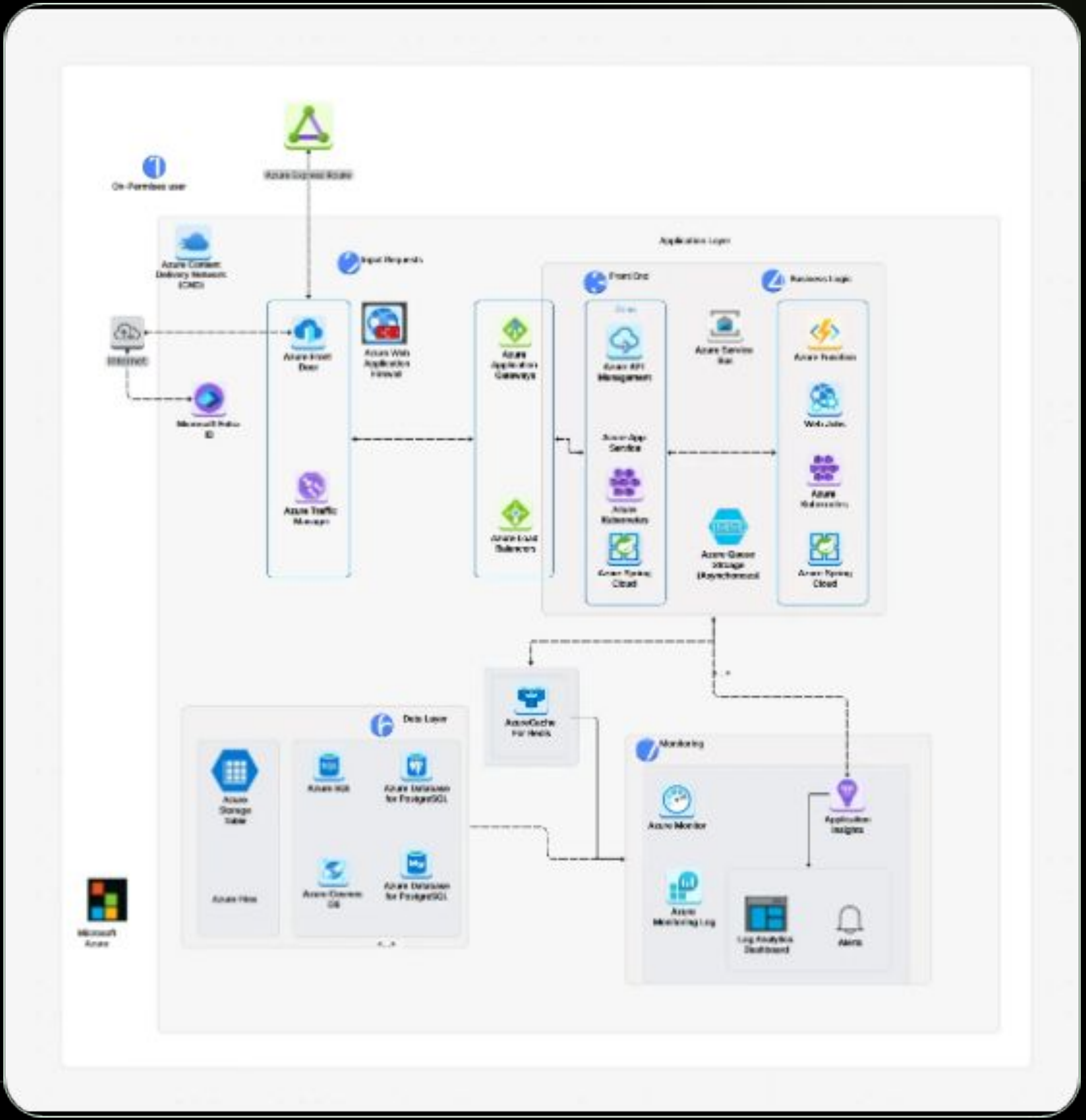
An automated, high-fidelity pipeline preserving architectural integrity while synthesizing professional furniture.

Anchor Preservation

YOLO-World (Zero-Shot)

Unlike standard detection, our first pass identifies architectural "anchors":

- **Structural Detection:** Windows, Fireplaces, and Doors.
- **Context Mapping:** Defining room limits and perspective.
- **Integrity:** Ensuring AI doesn't synthesize over windows.



Generative Synthesis

SeeDream 4.5 Engine

The core generative engine uses a state-of-the-art diffusion transformer architecture optimized for interior design.

It maintains lighting consistency and photorealistic textures, ensuring furniture looks integrated, not "pasted on."

Output: 2K High-Resolution photorealistic staging.

Dataset & Preprocessing

Source Data

5,000+ real estate images sourced from Zillow and Kaggle Open-Data.

Normalization

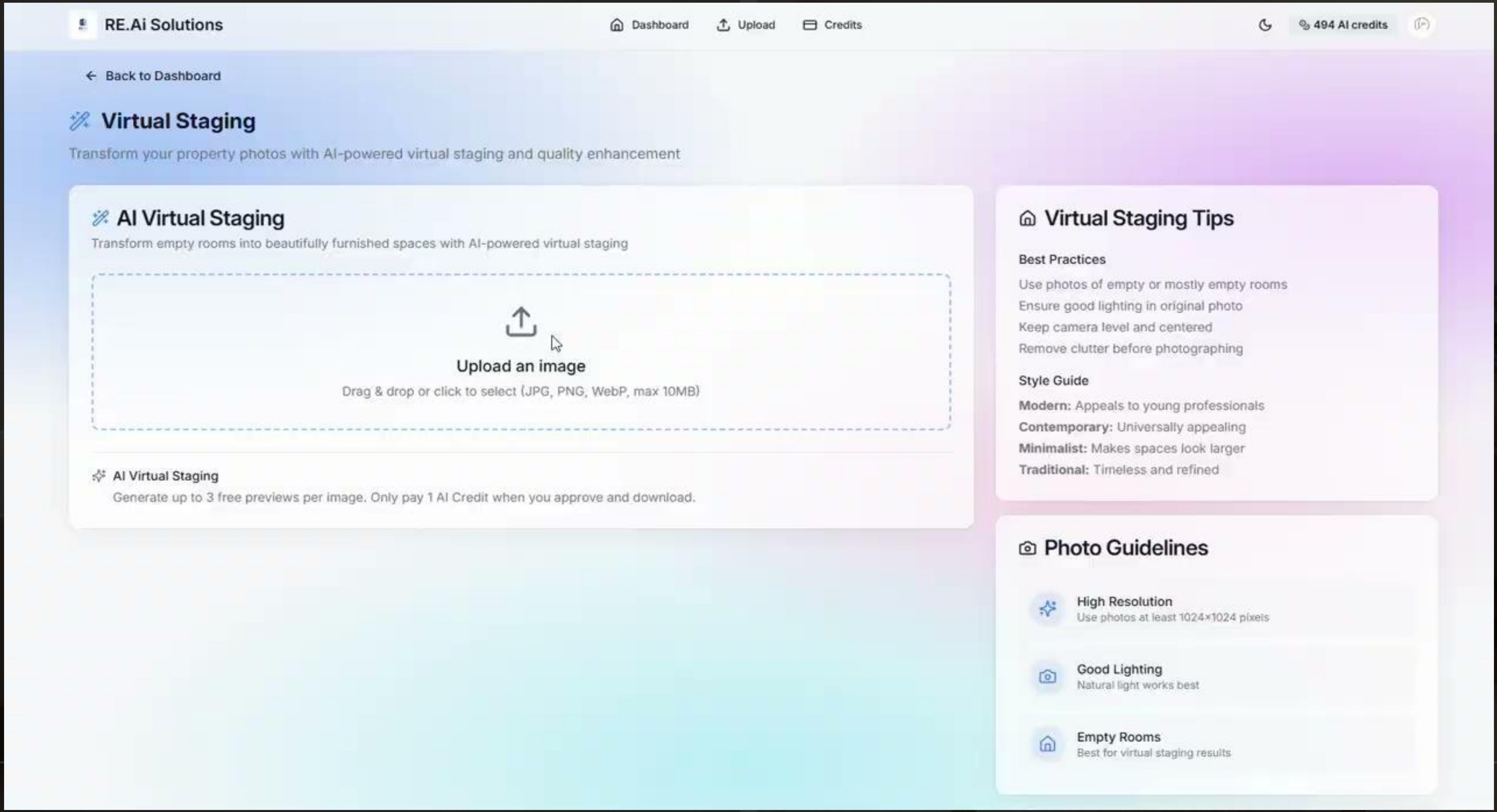
Images resized to 2048px; normalized lighting and color profiles.

Anchor Labels

Manual and automated labeling of room boundaries and structural anchors.

LIVE DEMO

Witness the transformation of a vacant space in real-time.



Key Performance Metrics

92.5%

Classification Accuracy

0.05s

Detection Latency per image

9.6 / 10

Structural Integrity Score

Efficiency Benchmark

RE.Ai reduces time-to-output by 95% compared to professional manual staging workflows.

Key Learnings

Vision Power

Zero-shot YOLO-World is superior for empty rooms where traditional context clues are absent.

Prompt Logic

Precise prompt engineering is critical to prevent AI from Hallucinating extra windows or doors.

Verification

The secondary YOLO pass (verification) significantly reduced spatial artifacts by 40%.

Roadmap to Production

Q1 2026

Mobile App Deployment (AR Integration)

Q2 2026

3D Matterport Tour Staging

Q3 2026

Full-Motion Video Staging

Q4 2026

Enterprise API Launch

Questions?

Thank you for your attention.

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https://github.com/BrandonLee-98/CV-Project-Virtual-Staging

Image Sources

https://cdn.cloudairy.com/blog/1773377983106Data%20Pipeline%20and%20App%20Architecture.webp

Source: cloudairy.com

https://images.unsplash.com/photo-1641527041985-9538bc8571ef?auto=format&fit=crop&w=328&h=264

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